Video as a multimedia information resource

- Video combines a changing sequence of visual frames with optional sound and text. It communicates time, actions, changes and events more directly than a still image.
- In a library, a short tutorial can demonstrate a catalogue search. In records management, a training video can show how an approved filing workflow works.
- A good video is planned for an audience, task, device and accessibility need; it is not merely footage with music.

KEY TAKEAWAY

A multimedia video succeeds when the intended user can understand and use the information.

Learning outcomes and expected evidence

- Explain frames, resolution, aspect ratio, frame rate, bitrate, codecs, containers and the difference between source and delivery files.
- Plan an ethical video using a clear purpose, script, storyboard, shot list, sound plan, permissions and accessibility provisions.
- Produce and evaluate a short edited clip plus captions, transcript, metadata, working files and a simple preservation plan.

KEY TAKEAWAY

The final work demonstrates both production technique and professional responsibility.

Why video is useful in BLISS and BRIM

- BLISS: orient new library users, show searching and borrowing, interpret exhibitions, and offer online visual demonstrations.
- BRIM: demonstrate safe records handling, interpret documented collections, explain records life-cycle procedures and present access-approved archive exhibitions.
- Both fields must distinguish persuasive edited video from an authentic evidential record; an edited summary must not be passed off as the complete source.

KEY TAKEAWAY

Use the same production tools, but adapt evidence, permissions and context to the profession.

The production life cycle

- Pre-production: identify users, objectives, content, rights, timeline, script, storyboard and locations.
- Production: capture image and sound, use a shot list, monitor quality, keep source files and record provenance.
- Post-production and release: edit a copy, add captions and other alternatives, export, review, publish, collect feedback and retain documentation.

KEY TAKEAWAY

Planning before recording prevents expensive retakes and ethical problems.

A meaningful learning objective

- Weak: “Make a library video.” This offers no defined task, user or success test.
- Stronger: “A first-year student will locate an e-book from the catalogue in under two minutes after viewing a 90-second tutorial.”
- The objective guides which screen actions must be visible, how much detail is needed and what to test with learners.

KEY TAKEAWAY

Define what the viewer should know or do after watching.

Storyboard, script and shot list

- A script specifies spoken words, on-screen text, transitions and accessibility cues. It is a content and timing plan rather than a decoration.
- A storyboard places visual sketches or descriptions in a sequence; each card names the scene, visual, narration, estimated duration and purpose.
- A shot list translates the storyboard into recording tasks: camera position, framing, props, needed permissions and sound requirements.

KEY TAKEAWAY

Storyboard before opening the editor; it makes missing information obvious.

Example storyboard: digital library orientation

- Scene 1 (0–6 s): title and purpose. Voice: “This guide shows how to locate a resource.”
- Scene 2 (6–12 s): the catalogue interface. Voice describes where to enter a title or subject.
- Scene 3 (12–18 s): results and availability. Scene 4 (18–24 s): next action and accessibility alternatives.

KEY TAKEAWAY

A four-scene example gives a testable sequence; the downloadable training clip follows it.

Frames and the illusion of motion

- A video frame is one still image in an ordered sequence. At playback, changing frames can create perceived movement.
- Frame rate is measured in frames per second (fps): a 10-second, 25 fps video contains about 250 displayed frames when rate is constant.
- Frame rate is not speed of playback. A 25 fps source can be played faster or slower by changing timing; screen recordings may also use variable rates.

KEY TAKEAWAY

Frame count ≈ duration × constant frame rate.

Choosing a frame rate

- Common use cases include 24 fps for a film-like aesthetic, 25/30 fps for general video, and 50/60 fps when smooth fast motion matters.
- Choose a rate matching equipment, lighting and delivery. Higher rates increase capture/data requirements at comparable quality settings.
- Do not claim that 60 fps is always “better”; motion style, bandwidth and production constraints matter.

KEY TAKEAWAY

Choose based on motion and delivery rather than maximum numbers.

Image dimensions and resolution

- 1920 × 1080 means 1,920 horizontal pixels and 1,080 vertical pixels per frame (about 2.07 million pixels).
- 1280 × 720 is about 0.92 million pixels per frame; 3840 × 2160 is about 8.29 million. More pixels can support detail if capture is sharp.
- Resolution alone cannot rescue an out-of-focus, poorly lit recording or unreadably small onscreen text.

KEY TAKEAWAY

Readability depends on source quality, display size and framing.

Aspect ratio and framing

- Aspect ratio compares display width to height. 16:9 is common for landscape tutorials and 9:16 for vertical short-form viewing.
- Cropping a 16:9 source to 9:16 can cut off faces, catalogue buttons, dates, document edges or other critical information.
- Protect titles and caption areas within a safe central composition; export and check on an actual target device.

KEY TAKEAWAY

Do not choose a platform crop that removes important evidence or instructions.

Shot types and camera angles

- Wide shot establishes location and context. Medium shot connects an action to a person or workspace. Close-up shows a specific detail.
- Example: establishing view of reading room → medium shot of librarian helping → close-up of catalogue result on screen.
- Use stable shots and intentional angles. A close crop should not silently omit details that change the meaning of a historical record.

KEY TAKEAWAY

Every shot should answer an information question.

Composition and continuity

- The rule of thirds is a helpful composition guide, not a law. Keep attention on the relevant action with appropriate contrast and uncluttered backgrounds.
- Continuity means successive shots preserve enough spatial and temporal information to remain understandable. Avoid confusing jumps in repeated gestures or object positions.
- For evidence or procedural training, insert clear labels where a scene is re-enacted, sped up or shown out of sequence.

KEY TAKEAWAY

Clarity and truthfulness are more important than cinematic effects.

Lighting, focus and stabilisation

- Use available diffuse light where possible; avoid positioning people against overly bright windows unless adjusted deliberately.
- Tap or manually set focus/exposure on the important subject. Keep the image stable using a tripod, support or controlled movement.
- Before a full recording, inspect a short test clip at normal viewing size for flicker, reflections, legibility and distractions.

KEY TAKEAWAY

A short test recording is one of the most useful quality controls.

Audio quality still matters

- Poor voice clarity often makes a visually attractive instructional video unusable. Record in a quiet place and monitor a test with headphones.
- Keep microphone placement consistent, manage ambient noise and avoid clipping. Do not use copyrighted music simply to fill silence.
- Provide captions and a transcript even when narration seems clear. Offer visual explanations of meaningful onscreen actions.

KEY TAKEAWAY

Viewers need to hear—or read—the information reliably.

Editing as a sequence of decisions

- Import source clips into an organised project; make trims on copies or in a non-destructive timeline; arrange scenes around the objective.
- Use cuts to remove irrelevant pauses and errors; add transitions only where they clarify a change. Check that graphics stay visible long enough to read.
- Export an access version, review from beginning to end and retain the source, project file and change log when required.

KEY TAKEAWAY

Editing should improve comprehension without hiding material facts.

The editing timeline

- Video tracks hold clips, titles and images; audio tracks hold narration, sound and other permitted sources. Tracks align along a time axis.
- Important techniques: trim, split, ripple edit, insert, crossfade, volume adjustment, simple colour correction and export.
- Timestamps, scene labels and file naming help reviewers locate decisions and reproduce outputs.

KEY TAKEAWAY

A timeline is a model of what appears and sounds at each moment.

Transitions, titles and on-screen text

- A direct cut is often enough. Excessive wipes, zooms and animations distract from an educational message.
- Use legible font sizes and high contrast; avoid covering key interface actions or metadata with captions or overlays.
- Include readable titles, names and dates only when accurate and authorised; identify reconstructions and simulated content clearly.

KEY TAKEAWAY

Decorative effects must never compromise readability or authenticity.

Animation: purpose and types

- Animation creates change from designed frames or keyframes rather than recording a continuous real-world action.
- 2D motion graphics can explain the lifecycle of a record; stop-motion demonstrates frame-by-frame movement; 3D animation can represent spaces or objects.
- Distinguish an illustrated reconstruction from authentic footage, especially in historical or archival contexts.

KEY TAKEAWAY

Use animation to explain an otherwise difficult process.

Keyframes, timing and easing

- A keyframe stores a property value at a particular time—for example, a title at x = 0 and then x = 600 pixels two seconds later.
- Software interpolates positions between keyframes. Linear movement has constant speed, while ease-in/ease-out changes speed gradually.
- For accessibility, avoid unnecessary flashing, allow users to pause motion and provide text or static alternatives where helpful.

KEY TAKEAWAY

Animation is controlled changes in properties over time, not merely moving decoration.

Motion graphics in a records lifecycle

- Example states: create/receive → classify → use/store → retain → transfer/dispose according to approved rules.
- Each visual stage can be animated with a labelled arrow, a time delay and a short narration describing the decision.
- Use accurate institutional retention schedules; do not let a simple graphic imply every record follows exactly the same route.

KEY TAKEAWAY

Animation can simplify a process while keeping the important decision points visible.

Captions, subtitles and transcripts

- Captions present speech and important non-speech sounds in sync with the video. They serve users who cannot hear audio and many other users.
- Translated subtitles present speech in another language. A transcript is a text document; descriptive transcripts can include meaningful visual information.
- Automatically generated captions may contain errors in names, accents and technical terminology; edit and verify them before release.

KEY TAKEAWAY

Captions must convey meaning, be timed, and be accurate.

Caption timing with WebVTT

- WebVTT is a common web caption format. A cue has a start timestamp, arrow and end timestamp, followed by caption text.
- Example: 00:00:06.000 --> 00:00:11.800 followed by “Enter the book title in the catalogue search box.”
- Check start < end, monotonic cue order, enough on-screen reading time, clear speaker changes and useful sound cues when needed.

KEY TAKEAWAY

A valid file structure is not sufficient: human readability and accuracy must also be checked.

Captions versus audio description

- Captions communicate needed audio. Audio description or integrated narration communicates important visual information for viewers who cannot see it.
- For a catalogue tutorial, narration should explain which field to select and what result appears; “click here” without context is inadequate.
- Use descriptions that are objective and concise, and provide a descriptive transcript when appropriate.

KEY TAKEAWAY

Accessibility concerns both the audio and the visual meaning.

Compression, container and codec

- A container such as MP4 holds video/audio streams and metadata. A codec such as H.264 encodes and decodes video; AAC is often used for audio.
- An MP4 file might use different codecs; the extension alone does not guarantee it will play on every device.
- Compression reduces size by removing redundancy or, for lossy codecs, some information; verify quality and playback after export.

KEY TAKEAWAY

Describe media by container, codecs, resolution, frame rate and bitrate.

Estimate compressed video storage

- Approximate file size (MB) = combined average bitrate in megabits/s × duration in seconds ÷ 8, ignoring small packaging overhead.
- Example: 3.0 Mb/s video + 0.128 Mb/s audio over 120 seconds ≈ 46.92 MB (decimal) or 44.75 MiB.
- Variable bitrate and metadata mean actual file sizes can differ; this calculation is an estimate, not a promise.

KEY TAKEAWAY

Video size depends strongly on combined bitrate and duration.

Preparing a low-bandwidth access copy

- For users with limited mobile data, use efficient encoding, readable images and audio, a sensible resolution and a text alternative.
- Export one tested access rendition; avoid repeatedly re-encoding the same compressed file because this can accumulate visible or audible damage.
- Make downloads intentional, display approximate file size, and check a phone/mobile connection when possible.

KEY TAKEAWAY

Availability of a video does not guarantee practical access.

Source, edit project and access derivative

- Source files are what the camera or recorder produced. The editing project records cuts, titles, sound and export settings.
- A viewing/access derivative is an output optimised for users. Its convenience does not make it the best preservation copy.
- Keep metadata, a simple change log, source integrity checks and any required access restrictions with the managed record.

KEY TAKEAWAY

Do not confuse a highly compressed access MP4 with the only archival copy.

Authenticity, provenance and consent

- Provenance identifies who created the recording, when, where, how and under which conditions. It is essential for evidential use.
- Consent to appear or be recorded is not always permission for public online distribution; records may contain sensitive information or third-party rights.
- For educational simulations, label staged scenes, reconstructions and AI-generated components; do not fabricate provenance.

KEY TAKEAWAY

Trust is supported by truthful context and traceable changes.

Basic video metadata record

- Descriptive: title, summary, subjects, creator, date, language and identifier.
- Technical: duration, resolution, fps or variable-rate note, container, video/audio codecs, approximate bitrate and file size.
- Administrative: rights owner, consent/permissions, access conditions, retention, version relationships and edit/export history.

KEY TAKEAWAY

A video without context is difficult to discover, reuse or preserve responsibly.

Quality assurance before publication

- Watch full video and check picture, sound, pacing, on-screen text, spelling, transitions, captions and timing.
- Confirm playback on intended devices, including keyboard/accessible controls and mobile bandwidth conditions.
- Ask a representative viewer to complete the intended task. Check that permissions, metadata, files and version labels are complete.

KEY TAKEAWAY

QA should include users and not only technical file playback.

Case study: BLISS library catalogue tutorial

- Need: new students cannot confidently locate reading materials using the online catalogue.
- Plan four scenes: overview; enter a title; interpret location/availability; next action. Include clear on-screen cues and captioned narration.
- Deliver: short MP4, edited VTT captions, transcript, source/project files, metadata and a brief usability test.

KEY TAKEAWAY

The product is successful if users can repeat the search independently.

Case study: BRIM archival exhibition video

- Need: visitors want to understand an approved historical collection without confusing illustrations with original records.
- Plan: verified description, approved stills or shots, source references, dates, narrator, visible labels for reenactments and restricted sections excluded.
- Deliver: access video, captions and descriptive transcript, metadata, rights inventory, version log and protected sources.

KEY TAKEAWAY

Historical storytelling needs attribution, context and editorial transparency.

Practical task: create a 60–90 second clip

- Choose BLISS or BRIM. Write one learning objective and a four-scene storyboard with time, camera/view and narration.
- Capture or assemble authorised original resources, make an edited clip, add accurate captions and provide a descriptive transcript.
- Submit the MP4, .vtt, transcript, storyboard, metadata, edit log and brief quality/permission reflection.

KEY TAKEAWAY

Production is a complete evidence-based workflow, not only a video file.

Video quality: review questions

- Can you justify frame rate, aspect ratio, recording choices, video formats, duration and approximate size?
- Can another person understand the video with captions and with useful descriptions of the visuals?
- Can you identify every source asset, permission, edit and exported copy? File formats, compression and delivery choices also affect access quality.

KEY TAKEAWAY

Plan → record → edit → caption → check → document → publish responsibly.